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SUBJECT:PYTHON
BATCH:03

Hangman Game in Python with Categories

Aim

To develop a Python-based Hangman game that allows a player to guess letters of a randomly selected word from different categories until they either guess the word correctly or run out of attempts.

Procedure

1. **Define word categories**: Create a dictionary with categories like `"Programming"`, `"Fruits"`, and `"Countries"`.
2. **Select a random word**: Use `random.choice()` to pick a category and a word from that category.
3. **Initialize game variables**: Set up a set to track guessed letters and define the number of allowed attempts (6).
4. **Display progress**: Show the word with underscores for unguessed letters using the `display()` function.
5. **Prompt user input**: Ask the player to guess one letter at a time.
6. **Validate guesses**:
   - Check if the letter was already guessed.
   - Deduct an attempt for wrong guesses.
7. **Check win condition**: If all letters of the word are guessed correctly, the player wins.
8. **End game**: If attempts reach 0, reveal the word and declare game over.

Code:

```
import random

# Word categories
words = {
    "Programming": ["python", "javascript", "compiler", "debugging"],
    "Fruits": ["apple", "banana", "orange", "mango"],
    "Countries": ["india", "canada", "germany", "brazil"]
}

def choose_word():
```

```
    category = random.choice(list(words.keys()))
    word = random.choice(words[category])
    return word, category

def display(word, guessed):
    return " ".join([letter if letter in guessed else "_" for letter in word])

def hangman():
    word, category = choose_word()
    guessed = set()
    attempts = 6

    print("\n🎯 Category:", category)

    while attempts > 0:
        print("\nWord:", display(word, guessed))
        print("Attempts left:", attempts)

        guess = input("Enter letter: ").lower()

        if guess in guessed:
            print("□ Already guessed!")
            continue

        guessed.add(guess)

        if guess not in word:
            attempts -= 1
            print("□ Wrong guess!")

        if all(letter in guessed for letter in word):
            print("\n🎉 You Won! Word was:", word)
            return

    print("\n💀 Game Over! Word was:", word)

hangman()
```

Output:

```
Output

🎯 Category: Fruits

Word: _ _ _ _ _ _
Attempts left: 6
Enter letter: b

Word: b _ _ _ _ _
Attempts left: 6
Enter letter: a

Word: b a _ a _ a
Attempts left: 6
Enter letter: n

🎉 You Won! Word was: banana

=== Code Execution Successful ===
```

Result

The game successfully selects a random word from a category and tracks letter guesses, showing progress and remaining attempts.

Conclusion

The Hangman game efficiently handles correct and incorrect guesses, providing an interactive and fun way to practice word guessing.